Logical Relational Model

Library Management System — EECS447: Database Systems
Fall 2026 · Professor Hossein Saiedian
Team Name: DataBASED

1. Introduction

Project Overview:

> This database will be a reliable library management system that efficiently manages library books, magazines, digital media, client memberships, borrowing and returning transactions, reservations, and fees. The system will help library staff manage library operations and allow clients to search for items, reserve books, and check their loan status on their library membership. The database will also provide useful reports for both users for library fines, book availability, client activity, and more.

Scope:

> The library management system will manage books, digital media, and magazines; client accounts and membership types; borrowing and return transactions; reservations; and notifications for due dates, overdue items, and available reservations. Membership types will determine borrowing limits and fee rules. The system will generate reports on fines, item availability, borrowing trends, and client activity. It will include a staff interface for managing items, loans, returns, and accounts, and a client interface for searching the catalog, reserving items, and checking loan status. Work outside these library collection, circulation, and account functions is beyond the scope of this project.

2. Relational Schema Mapping

- Identify relations

ER component	Type	Relation
Item (superclass)	Entity	ITEM
Book	Entity (subclass of Item)	BOOK
Digital Media	Entity (subclass of Item)	DIGITAL MEDIA
Magazine	Entity (subclass of Item)	MAGAZINE
Membership	Entity	MEMBERSHIP
User (Client)	Entity	CLIENT
Loan	Relational entity	LOAN
Reservation	Relational entity	RESERVATION
ISA	Specialization	None; each subclass share the

		PK,FK itemID from ITEM
Has (Membership to Client)	Relationship	None; MembershipType is a FK in CLIENT
Has (Loan to Item)	Relationship	None; itemID is a FK in Loan
Has (Reservation to Item)	Relationship	None; itemID is a FK in Reservation
Makes (Client to Loan)	Relationship	None; ClientID is a FK in Loan
Makes (Client to Reservation)	Relationship	None; ClientID is a FK in Reservation

- Define attributes and domains

ITEM

Attributes	Type	Domain/Constraints	Description
ItemID	INT	PK, positive, and incremental integer	Unique identifier assigned to each item
Title	VARCHAR(255)	Not `NULL`	Title of item
AvailabilityStatus	VARCHAR(20)	Not `NULL`; ‘Available’ or ‘Checked out’	Whether the item can currently be borrowed
Restricted	VARCHAR(20)	Not `NULL`; ‘None’ or ‘Restricted’	Whether borrowing of the item is restricted

BOOK

Attributes	Type	Constraints	Description
ItemID	INT	PK, FK → ITEM(ItemID)	Unique identifier assigned to each item; References the PK from ITEM
Author	VARCHAR(100)	Not `NULL`	Author of the book
Genre	VARCHAR(50)	Not `NULL`	Genre of the item
PubYear	YEAR	N/A	Year the book was

			published
ISBN	CHAR(13)	Not ‘NULL’, Unique	International Standard Book Number
Format	VARCHAR(30)	N/A	Format of the book
Publisher	VARCHAR(20)	N/A	Company that published the book

DIGITAL MEDIA

Attributes	Type	Constraints	Description
ItemID	INT	PK, FK → ITEM(ItemID)	Unique identifier assigned to each item; References the PK from ITEM
Creator	VARCHAR(100)	Not `NULL`	Creator of the media
Format	VARCHAR(30)	N/A	Format of item (movie, music, tv show etc.)
Genre	VARCHAR(50)	Not `NULL`	Genre of the item
ReleaseYear	YEAR	N/A	Year item was released

MAGAZINE

Attributes	Type	Constraints	Description
ItemID	INT	PK, FK → ITEM(ItemID)	Unique identifier assigned to each item; References the PK from ITEM
IssueNum	VARCHAR(20)	N/A	Issue number of magazine
PubDate	DATE	N/A	Date magazine was published
Publisher	VARCHAR(20)	N/A	Company that published the magazine

MEMBERSHIP

Attributes	Type	Constraints	Description
MembershipType	VARCHAR(10)	Unique and must be ‘Regular’, ‘Student’, or ‘Seniors’	Type of memberships available
BorrowingLimit	INT	Not ‘Null’	Max amount of items member can borrow
DailyFine	DECIMAL(5,2)	Between 3 and 8	Amount of money charged for late returns
LoanPeriod	INT	Not ‘NULL’	Amount of days member can borrow items for
ReserveLimit	INT	Not ‘NULL’	Max amount of items member can reserve

CLIENT

Attributes	Type	Constraints	Description
ClientID	INT	Positive, unique, and incremental integer	Unique client identifier
FirstName	VARCHAR(50)	Not ‘NULL’	First name of client
LastName	VARCHAR(50)	Not ‘NULL’	Last name of client
Email	VARCHAR(250)	N/A	Email address of client
AccountStatus	VARCHAR(20)	“Active” or “Suspended”	Status of client’s account
FeesOwed	DECIMAL(5,2)	N/A	Total amount of fees owed by client
MemberType	VARCHAR(10)	FK → MEMBERSHIP(MembershipType)	Membership type of client

LOAN

Attributes	Type	Constraints	Description
LoanID	INT	PK, positive, and incremental integer	Identifier of loan

ClientID	INT	FK → CLIENT(ClientID), Not NULL	Identifier of client
ItemID	INT	FK → ITEM(ItemID), Not NULL	Identifier of Item
CheckoutDate	DATE	Not 'NULL'	Date items checked out
DueDate	DATE	Not 'NULL'	Date items due back
ReturnDate	DATE	N/A	Date items returned by client
FineAmount	FLOAT	N/A	Late fees accrued from loan

RESERVATION

Attributes	Type	Constraints	Description
ReservationID	FLOAT	PK, positive, and incremental integer	Identifier of reservation
ClientID	FLOAT	FK → CLIENT(ClientID), Not NULL	Identifier of client
ItemID	FLOAT	FK → ITEM(ItemID), Not NULL	Identifier of item
ReservationDate	DATE	Not 'Null'	Date reservation was made
ReservationStatus	VARCHAR(20)	Not NULL; "Pending", "Ready", "Expired", or "Fulfilled"	Status of reserved item

- Determine primary keys

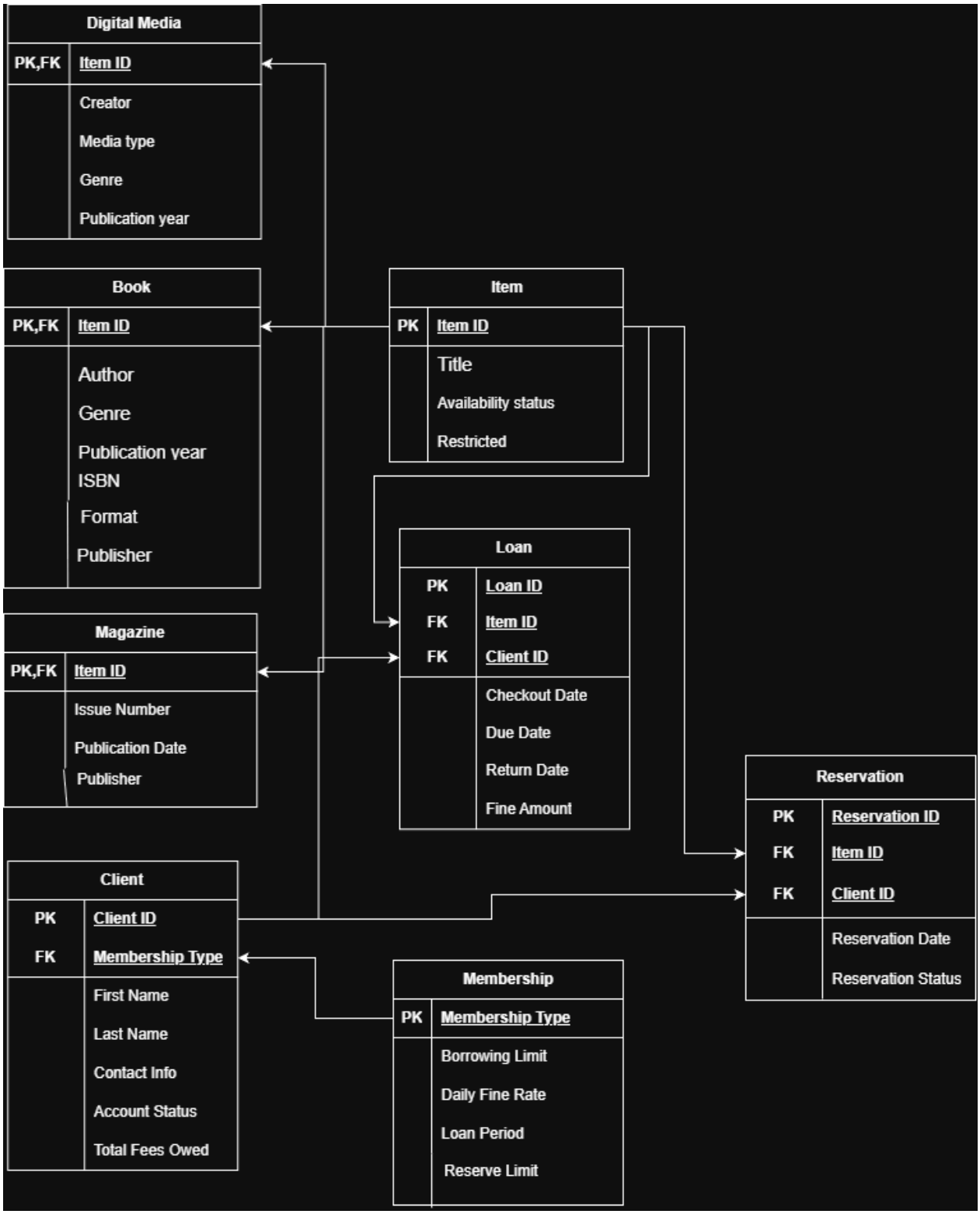
Relation	Primary Key
Item	ItemID
Book	ItemID
Digital Media	ItemID
Magazine	ItemID

Membership	MembershipType
Client	ClientID
Loan	LoanID
Reservation	ReservationID

- Establish foreign keys

Relation	Foreign Key	References
Book	ItemID	Item(ItemID)
Digital Media	ItemID	Item(ItemID)
Magazine	ItemID	Item(ItemID)
Client	MembershipType	Membership(MembershipType)
Loan	ClientID	Client(ClientID)
Loan	ItemID	Item(ItemID)
Reservation	ClientID	Client(ClientID)
Reservation	ItemID	Item(ItemID)

3. Relational schema diagram
https://drive.google.com/drive/folders/1chZW5eF8LOi55M-m4x8tVznibE8yhCb1?usp=sharing



4. Schema Documentation

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